

# Dhairya Parekh

Data Scientist / Machine Learning Engineer

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## SUMMARY

Experienced Data Scientist and Machine Learning Engineer with over 4 years of hands-on expertise in developing and deploying predictive models to enhance fraud detection, customer segmentation, and operational efficiency. Adept at collaborating with cross-functional teams to identify data-driven insights that inform decision-making. Proficient in Python, SQL, and machine learning frameworks, with a deep understanding of A/B testing, data visualization, and advanced analytics techniques. Driven by a passion for leveraging advanced machine learning techniques and strategic leadership to deliver impactful, data-driven solutions that transform business outcomes.

## SKILLS

**Tools and Technical skills:** Python, R, C, C++, Java, SQL, MySQL, PostgreSQL, SQL Server, Regression, Classification, SVM, Decision Trees, Random Forests, Naive Bayes, KNN, CNN, RNN, LSTM, Transformers, BERT, GPT, Statistical Analysis, Hypothesis Establishment, A/B testing, TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, SciPy, Seaborn, Ggplot2, Flask, Django, Tableau, Power BI, Visual Studio Code, PyCharm, Jupyter, Git, AWS (S3, Lambda, EC2, SageMaker), Azure, GCP

## EXPERIENCE

### CitiGroup, TX | March 2023 - Current | Data Scientist / Machine Learning Engineer

- Designed and implemented a fraud detection model utilizing XGBoost on a dataset of over 1 million credit card transactions with 350+ features, optimizing memory usage by 30% and enhancing computational performance by 40%.
- Engineered critical features such as transaction deviations, product fraud likelihoods, and temporal patterns, leveraging Principal Component Analysis to improve model accuracy and reduce training time by 20%.
- Conducted comprehensive analyses of fraud trends across various devices, card types, and email domains, identifying high-risk patterns and collaborating with fraud prevention teams to develop cumulative capture rate visualizations for stakeholders.
- Achieved an AUC score of 0.91 and a concordance score of 0.97, successfully capturing 75% of fraudulent transactions in the top 10% of probability bins, contributing to potential annual savings of millions for the organization.
- Refined model probability outputs using calibration curves, effectively aligning predictions with actual fraud occurrences, reducing false positives, and significantly improving overall detection precision.
- Collaborated closely with cross-functional teams to optimize fraud detection strategies, utilizing insights gained from data analyses to inform decision-making and enhance operational effectiveness in fraud prevention efforts.

### Adons Softech, India | June 2018 – June 2021 | Data Scientist / Machine Learning Engineer

- Developed and deployed deep learning models using ConvLSTM to detect real-time electrical appliance consumption, achieving 93% classification accuracy and 89% regression accuracy.
- Automated data collection processes and conducted exploratory data analysis, reducing data acquisition costs by 90%.
- Designed and implemented ML architecture for real-time and batch time-series data processing with a throughput of 100 data points per second.
- Forecasted user energy consumption patterns with 92% accuracy on a daily and monthly basis, performing cluster analysis to enhance user engagement by 20%.
- Segmented 1.6 million customers using K-means clustering and inertia analysis to identify optimal clusters, improving marketing strategies.
- Developed Tableau dashboards and conducted A/B testing on 550k users, leading to an 18% increase in marketing responses and significant improvements in ad targeting.

## EDUCATION

**Master of Science in Computer Science** | University of Texas at Arlington, Arlington, TX

**Bachelor of Engineer in Computer Engineering** | L.D. College of Engineering, Ahmedabad, Gujarat, India

## CERTIFICATION

**Finetuning Large Language Models, LLMOps, Quality and Safety for LLM Applications by Andrew Ng | Dec 2023**